

and group them. Online Appendix Table A3 summarizes the relationship between groups and ICD-10 codes.

We also use several data sources to assess the effect of CAPS on public spending. One limitation here is that spending is divided between federal and local governments. The spending data at these levels are aggregated in different ways, which can prevent accurate comparisons across them. With this caveat in mind, we use data on federal spending on hospital admissions from the Ministry of Health's SIH (DATASUS 1979–2024e) and on municipal health spending from Brazil's National Treasury (Instituto de Pesquisas Econômica Aplicada 2023b).

For mortality outcomes, we use data from the Mortality Information System (*Sistema de Informação sobre Mortalidade*—SIM) (DATASUS 1979–2024c). SIM records all deaths in the country, with their causes (coded using the ICD-10). We focus on deaths related to alcohol, overdose, suicide, and the larger category of deaths of despair, which is the aggregation of the other three (Case and Deaton 2015, 2017). In addition, we gather data on homicides, which can be a proxy for violent crimes more generally (Dix-Carneiro, Soares, and Ulyssea 2018). We also investigate overall mortality and infant mortality as placebo outcomes. To measure infant mortality, we collect data on the number of children born alive at each municipality from the Ministry of Health's registers (DATASUS 1979–2024b).

To control for differential trends in important determinants of mental health in our empirical strategy, we use several sources of data on municipality characteristics. The Brazilian Statistical Office (*Instituto Brasileiro de Geografia e Estatística*—IBGE) estimates each municipality's GDP, from 2002 to 2016 (Instituto Brasileiro de Geografia e Estatística 2002–2021). The Ministry of Social Development (MDS/SAGI) provides data on the main conditional cash transfer policy in Brazil, the *Bolsa Família* program (PBF) (Instituto de Pesquisas Econômica Aplicada 2023a). The Brazilian Ministry of Health records data on the age and gender composition of people in each municipality (DATASUS 1980–2021). The IPEA provides a set of geographical variables from the municipalities in the baseline (Instituto de Pesquisas Econômicas e Aplicadas 1995–2024). We also gather information on the number of mental health providers and offices in the baseline from the IBGE's 2002 Health and Medical Care Survey (*Pesquisa de Assistência Médico-Sanitária*—AMS) (Instituto Brasileiro de Geografia e Estatística 2002). Finally, we use the results of the 2000 Brazilian Census for data on socioeconomic variables (Instituto Brasileiro de Geografia e Estatística 2000).

Merging all the data described above, we end up with a sample of balanced yearly data for 5,180 municipalities covering all years between 2002 and 2016. Online Appendix Table A2 provides summary statistics.

III. Study Design and Estimation Strategy

We exploit the staggered implementation of CAPS starting after 2002 within a DID framework to analyze the effects of this intervention on outcomes related to public mental health. In such a setting, traditional two-way fixed effects estimators are biased if treatment effects are heterogeneous over time or across cohorts (Goodman-Bacon 2021; Sun and Abraham 2021). To circumvent this problem, we

follow De Chaisemartin and D'Haultfoeuille (2020) and estimate well-defined and relevant causal parameters, robust to treatment effect heterogeneity.¹⁰

In summary, for any period t when a group of municipalities receives CAPS, we can use standard 2×2 DID to get an estimate for that group. For a fixed event-time $k \geq 0$, this is the result of comparing across periods $t - 1$ and $t + k$ units that switched their status from untreated to treated at t with those remaining untreated until $t + k$. We then aggregate these separate DID estimates based on the size of each treated cohort to estimate the average effects for all groups of treated municipalities, k periods away from the year they received CAPS. Online Appendix Section B formalizes this strategy. Regarding inference, we compute standard errors using a municipality-level clustered bootstrap.

Our research design uses groups whose treatment is stable, to infer the trends that would have affected switchers if they had not gained a CAPS. The only way that this design would be invalid would be if trends in time-varying determinants of outcomes across switchers and nonswitchers diverged for some reason other than the CAPS. To deal with such concerns, our main specification includes state-specific nonparametric trends. These seem particularly relevant in Brazil since many public policies—such as those related to education and public security—are largely determined at the state level.

As we show next, disadvantaged areas were less likely to construct a CAPS than areas with more resources. Although these differences tend to be small in magnitude, a risk exists that differential trends (driven by these baseline differences) spuriously correlate with the treatment effect. This could occur if disadvantaged and advantaged areas had already started on different paths, in terms of the outcomes of interest, before the psychiatric reform. To account for the possibility of this threat to internal validity, we control for interactions of a wide range of pretreatment municipality characteristics with a linear time trend: Theil index, poverty rate, illiteracy rate, the share of population living in rural areas, population, social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to the capital, temperature, and rainfall (Carrillo and Feres 2019).

We also control for differential trends according to temporal variation in potential confounders. During the 2000s, Brazil experienced high economic growth and demographic changes. Therefore, we control for local economic conditions measured by GDP per capita and the age-by-gender composition of municipal population (the share of inhabitants within each nine-year-by-gender bracket, from 10–19 up to 79 years). We also consider more flexible trends according to annual per capita PBF expenditure, as the Brazilian cash transfer program started expanding across municipalities in 2003. Reassuringly, point estimates are stable to different specifications, suggesting that our results are unlikely to be driven by differential trends across switchers and nonswitchers. Section V discusses whether other policy changes could confound our results.

To further assess the plausibility of our identification assumption, we follow Galiani, Gertler, and Schargrodsky (2005) and Rocha and Soares (2010) and estimate a hazard

¹⁰In such a staggered design, the estimators we use are very similar to those proposed by Sun and Abraham (2021) and Callaway and Sant'Anna (2021) (for this particular case, in a specification without covariates).

model of the probability that a municipality gains a CAPS.¹¹ Results are reported in online Appendix Table A4, with marginal effects calculated on independent variables averages. Overall, we find that some baseline characteristics are correlated with the probability of receiving a CAPS. Still, the effects are quantitatively small: the effects of one standard deviation are smaller than 2 percentage points in all cases. Furthermore, and fundamentally for our identification assumption, we see no correlation between receiving a CAPS and the pretreatment dynamic of mental health, crime, and economic indicators. These results support the validity of our empirical strategy.

We can also estimate treatment effects in pre-CAPS periods to test the plausibility of the underlying parallel trends assumption defining our DID design. Following De Chaisemartin and D'Haultfoeuille (2020) and Callaway and Sant'Anna (2021), we use a placebo estimator that replaces the "long comparisons" across $t - 1$ and $t + k$ (between groups that switched from untreated to treated at t and those remaining untreated until $t + k$) with "short comparisons" across $t' - 1$ and t' for all $t' < t$. Different from TWFE regressions, we do not have a universal baseline period. For each placebo treatment period t' , we compare it to the immediately preceding period. In this case, our pretreatment parameters are pseudo-ATTs: they are the treatment effects we would have estimated had the treatment taken place at the placebo date.¹² Finding coefficients statistically different from zero would indicate that the parallel trends assumption is violated. The placebo effects are generally statistically indistinguishable from zero for our primary outcomes, again bolstering our design validity. In the few cases with significant pretreatment effects, these are close to zero and have much lower magnitude than the post-treatment effects, making it unlikely that the existing slight variation across groups explains the treatment effects.

IV. Main Results

We present our main results in graphical form, plotting together in one figure dynamic effects, placebo effects, and their 95 percent confidence intervals. In the text, we report the standard errors in parentheses. We group our results into five sets, each connected to one element of the diagram in Figure 2.

A. Mental Health Care Supply

The effects of Brazil's psychiatric reform on the supply of mental health practitioners are presented in Figure 3 and Table 1. Our estimates indicate a remarkable

¹¹ We perform this estimation by modifying our dataset, so each municipality leaves the sample after getting a CAPS. Then, we estimate logit models controlling for a flexible polynomial of time, where the dependent variable is a dummy indicating that a municipality gained a CAPS and the independent variables are socioeconomic variables. We estimate three models considering the first, second, and third lags of the changes in our variables of interest. We additionally include some baseline characteristics as independent variables.

¹² In this case, all our event-study estimates have a similar interpretation. In TWFE regressions, one has to normalize relative to a universal baseline period (generally the period immediately before the treatment starts); otherwise, parameters are not identified due to perfect multicollinearity. Under this strategy, event study estimates in pretreatment periods are not treatment effect parameters, so they differ to our pretreatment estimates in terms of interpretation. Anyway, either case is just a linear combination of the other, so both are valid strategies to pretest the parallel trends assumption.

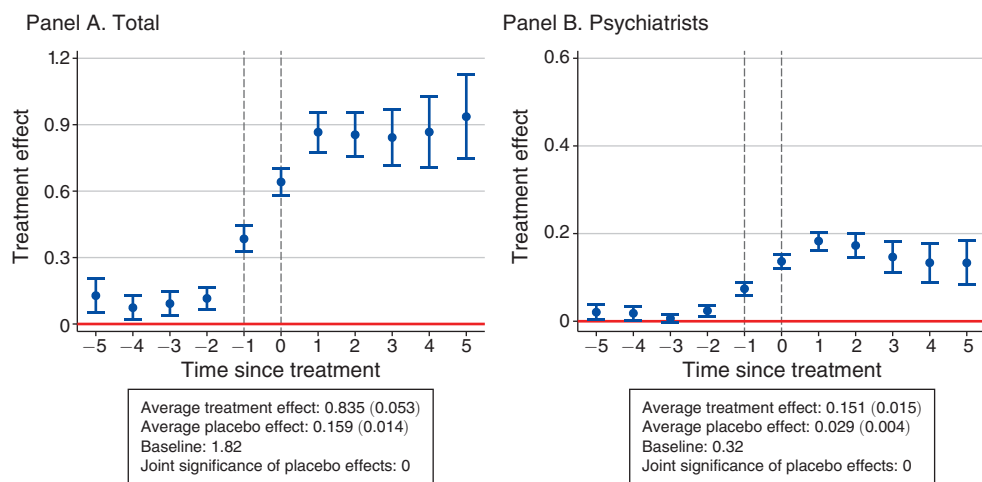


FIGURE 3. EFFECTS OF CAPS ON MENTAL HEALTH PRACTITIONERS, PRACTITIONERS PER 10,000 PEOPLE

(continued)

increase in the supply of professionals: the average treatment effect indicates an increase of 0.84 professionals per 10,000 people (46 percent over the average of the year before the event,¹³ s.e., 0.05). The increase precedes the CAPS' introduction, and reaches its peak after the policy took effect. For psychiatrists, the treatment effects indicate a statically significant increase of 0.18 professionals per 10,000 people (60 percent, s.e., 0.01) one year after municipalities receive a CAPS. Treatment effects decay monotonically after that, reaching 0.13 (s.e., 0.02) points five years after treatment.

According to the law regulating CAPS implementation, if smaller municipalities fail to hire psychiatrists, they can hire general practitioners and family doctors. We estimate the policy effects on the supply of these professionals in Table 1 and online Appendix Figure A4. The point estimates are generally positive, albeit imprecise, and have a much lower magnitude than those for psychiatrists. We also look at the effects on the overall supply of physicians, which is positive and increases over time. It can be accounted for mainly by the increased supply of psychiatrists, suggesting no substitution across specialties.

In the same figure and table, we also show estimates for the effect of CAPS on other mental health providers that usually constitute community mental health teams: psychologists, occupational therapists, and social workers. Overall, the pattern is similar to the one for psychiatrists, and CAPS' effects on the supply of these professionals are significant and high in magnitude. As before, there are anticipation effects, most marked in the year before the establishment of CAPS. Treatment effects then rise after the beginning of the intervention. The number of psychologists per 10,000 inhabitants increased by 0.22 (27 percent, s.e., 0.02) in

¹³ Henceforth, percentages indicate the size of the effect in percentage terms compared with the average of each outcome in the year before the event, unless otherwise noted.